ADMP Project

Group 15

SHU CONSULTANCY 2020-2024



Credit: Fsm.ac.in

Meet Group 15



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Overview of the Road Traffic Accident Dataset



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Road traffic accidents (RTAs) in the UK remain a primary driver of motor insurance risk, claims volatility, and pricing complexity.

"Since 1979, there has been a general downward trend in the number of people killed on roads in Great Britain with a flatter trend in the decade since 2010. In 2024, road casualties showed a slight decline from 2023." Department for Transport (2024).

RTAs contribute significantly to economic and insured losses, with road crash costs estimated at 2–3% of GDP in developed economies (Elvik et al., 2016). For insurers, this translates into substantial reserve requirements, capital allocation pressures, and increased profitability risks.

Our Client ABC Insurance has asked us to investigate the effect of RTAs over the past 5 years from the gov.uk website and provide data driven recommendations to support more accurate risk assessment, optimise premium pricing, improve claims cost forecasting, and enhance overall profitability.

Overview of the RTA Business Questions

1. Provide a monthly breakdown of the number of collisions by collision severity, vehicle type, vehicle age and urban vs. rural areas in the UK from 2020 to 2024.

Research has shown that road traffic accidents (RTA) severity is influenced by multiple factors, such as vehicle characteristics, environmental conditions, and spatial context. Foundational studies highlight that severity is strongly associated with variables such as road conditions, driver behavior, and vehicle-related factors (Savolainen et al., 2011).

2. Provide a monthly breakdown of the number of collisions by collision severity, driver age group, sex of driver & light conditions in the UK between 2020 to 2024.

Driver demographics particularly age and gender, which affect both driving exposure and behavioural risk. Evidence shows that younger drivers have higher crash involvement rates due to inexperience, while older drivers may face increased injury severity due to physical vulnerability (McCartt et al., 2009).
Gender differences are also well established, with male drivers more frequently involved in high severity collisions, reflecting differences in driving patterns and risk-taking behaviour (Ozkan & Lajunen, 2006).
Environmental factors, particularly lighting conditions, further influence accident severity. Reduced visibility in darkness has been shown to significantly increase both crash likelihood and fatality risk, especially in rural or poorly lit areas (David et al., 2019).

3. What were the top ten worst vehicle make & models by the number of fatal collisions across urban and rural areas for each month in the UK between 2020-2024

Vehicle analysis is critical in understanding fatal road traffic accidents, as vehicle characteristics and drivers interactions with vehicle techs significantly influence crash outcomes. Research shows that factors such as vehicle type, performance, and structural design contribute to variations in fatality risk, supporting the need for granular analysis at the level of vehicle make and model (Savolainen et al., 2011).
In 2024, the majority of road fatalities occurred on roads in rural areas, whereas the majority of all casualties occurred on roads in urban areas. Department for Transport (2024). It is necessary that this is investigated over a 5 year period.

Overview of the RTA Business Questions

4. Provide a count of the number of collisions by collision severity, engine capacity, light conditions and month for cars between 2020-2024 in urban areas of the UK

Vehicle size often categorized by engine capacity is a critical determinant of collision risk and severity. Research indicates that variations in vehicle mass and structural characteristics influence both crash likelihood and injury outcomes, with heavier or larger vehicles exhibiting different severity dynamics compared to smaller vehicles (Wenzel, 2016).
Moreover, crash severity datasets highlight the importance of integrating variables such as light conditions, urban classification to capture variations in accident patterns (Yang et al., 2024)

5. Provide a count of the number of collisions by collision severity, work & personal use vehicles, sex & age group of driver and month from 2020 - 2024 in the UK

Temporal variation further reinforces the need for a monthly analytical approach, as accident frequency and severity fluctuate over time due to changes in driving exposure, behaviour, and environmental conditions. Studies show that crash risk varies across age, gender, and time-of-day or temporal factors, indicating that risk is dynamic rather than static (Brian C Tefft, 2018)



Key Features of Dataset & Integration



Credit: Freepik.com

Collision dataset has **503,475** observations **and 44** variables
Vehicle dataset has **920,692** observations **and 32** variables
Casualty dataset: **640,522** observations **and 23** variables

Three related datasets were used: Collision, Vehicle, and Casualty datasets from gov.uk

Dataset was from the last 5 years of Road Traffic Accidents from 2020 – 2024

collision_index was used as the main key to link the datasets.

Collision data provides various attributes related to time and location of each accidents.

Vehicle data provides information on vehicles involved in each recorded collision.

Casualty data provides information about individuals injured or dead in road traffic collisions.

The data was imported directly from the RTA gov.uk website with an R function "fread" to ensure automation and easy access

Key Features of Dataset & Integration

Collision dataset

```
##    collision_index collision_year collision_ref_no location_easting_osgr
##             <char>          <int>           <char>                 <int>
## 1:   2021170H10421           2021         170H10421                447098
## 2:   2021170H11231           2021         170H11231                450486
## 3:   2020170M11750           2020         170M11750                449694
## 4:   2021170M31761           2021         170M31761                449744
## 5:   2021170S10441           2021         170S10441                445971
## 6:   2022170S10952           2022         170S10952                443914
##    location_northing_osgr longitude latitude police_force collision_severity
##                     <int>     <num>    <num>        <int>              <int>
## 1:                 532997 -1.270905 54.68983           17                  3
## 2:                 533118 -1.218333 54.69059           17                  3
## 3:                 519733 -1.232884 54.57040           17                  3
## 4:                 514217 -1.233040 54.52083           17                  3
## 5:                 520834 -1.290292 54.58064           17                  3
## 6:                 521309 -1.322030 54.58508           17                  3
##    number_of_vehicles number_of_casualties       date day_of_week   time
##                 <int>                <int>     <char>       <int> <char>
## 1:                  2                    2 22/05/2021           7  22:44
## 2:                  2                    1 20/10/2021           4  15:50
## 3:                  2                    1 01/12/2020           3  18:00
## 4:                  1                    1 09/12/2021           5  16:55
## 5:                  3                    1 12/04/2021           2  09:02
## 6:                  2                    1 05/07/2022           3  17:00
```

Vehicle dataset

```
##    collision_index collision_year collision_ref_no vehicle_reference
##             <char>          <int>           <char>              <int>
## 1:   2020210979534           2020        210979534                  2
## 2:   2020030959800           2020        030959800                  1
## 3:   2020360978858           2020        360978858                  2
```

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```
## 4:   2020430340711           2020        430340711                  1
## 5:   2020010260297           2020        010260297                  1
## 6:   2020010285121           2020        010285121                  2
##    vehicle_type towing_and_articulation vehicle_manoeuvre_historic
##           <int>                  <int>                      <int>
## 1:            5                      0                          3
## 2:            5                      0                         17
## 3:            5                      0                         18
## 4:            5                      0                         18
## 5:            5                      0                         99
## 6:            4                      0                         18
```

Casualty dataset

```
##    collision_index collision_year collision_ref_no vehicle_reference
##             <char>          <int>           <char>              <int>
## 1:   2020010280094           2020         010280094                  1
## 2:   202031D109620           2020         31D109620                  1
## 3:   2020401003715           2020         401003715                  1
## 4:   2021201086330           2021         201086330                  1
## 5:   2021371050963           2021         371050963                  1
## 6:   2022161255871           2022         161255871                  1
##    casualty_reference casualty_class sex_of_casualty age_of_casualty
##                 <int>          <int>           <int>           <int>
## 1:                  1              3               2              24
## 2:                  1              3               2              95
## 3:                  1              3               1              39
## 4:                  1              3               1              63
## 5:                  1              3               1              51
## 6:                  1              3               2              60
##    age_band_of_casualty casualty_severity pedestrian_location
##                   <int>             <int>               <int>
## 1:                    5                 3                   8
## 2:                   11                 2                   5
## 3:                    7                 3                   5
## 4:                    9                 3                   6
## 5:                    8                 3                   9
## 6:                    9                 3                   8
##    pedestrian_movement car_passenger bus_or_coach_passenger
##                  <int>         <int>                 <int>
## 1:                   9             0                     0
## 2:                   1             0                     0
## 3:                   1             0                     0
## 4:                   9             0                     0
## 5:                   9             0                     0
## 6:                   3             0                     0
```

The extracted columns to be used in the analysis are:

- collision_index
- collision_severity
- collision_date.
- light_conditions.
- urban_or_rural_area
- vehicle_type
- sex_of_driver
- age_band_of_driver
- age_of_vehicle
- generic_make_model
- vehicle_reference
- casualty_reference
- area casualty_severity

Two variables were also created:

- work_or_personal: this was to explain the type of vehicle involved in the collision, whether work or personal vehicle
- engine_capacity_bands: this grouped the engine capacity into bands

Overview of Software Solution Used



Apache Hive and Apache Hadoop was used to store, process, analyse, query, and manage the datasets used.



R studio has been used to extract the data from the web and used in cleaning and exporting as Parquet.



DbSchema

DbSchema was used as the database design tool.

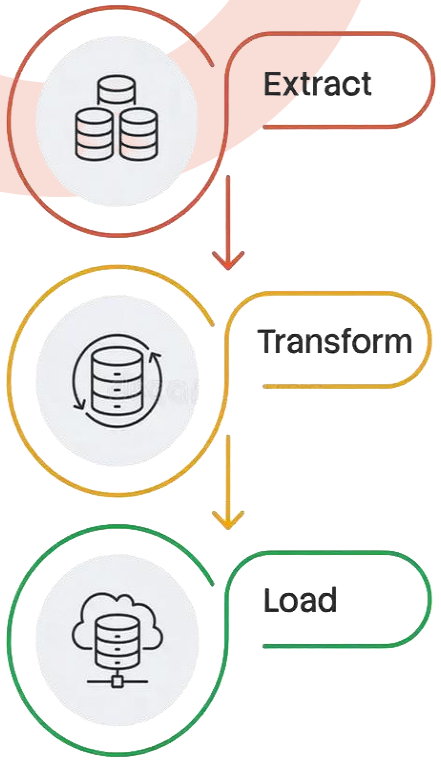


Tableau connects to hive and has been used in creating the charts to answer each business question.



Microsoft PowerPoint has been used in presenting observations to stakeholders

Overview of the Data Warehouse Development



ETL (Extract, Transform, Load) approach was used because it is assumed that the UK road safety data contains possible missing values, inconsistent encoding for the variables and formatting issues. By using ETL, only relevant, cleaned, and structured data is loaded into HDFS/Hive. This in turn reduces storage usage and ensures faster query processing time.

Data was extracted from correlated datasets via gov.uk:

- Collision Dataset
- Vehicle Dataset
- Casualty Dataset

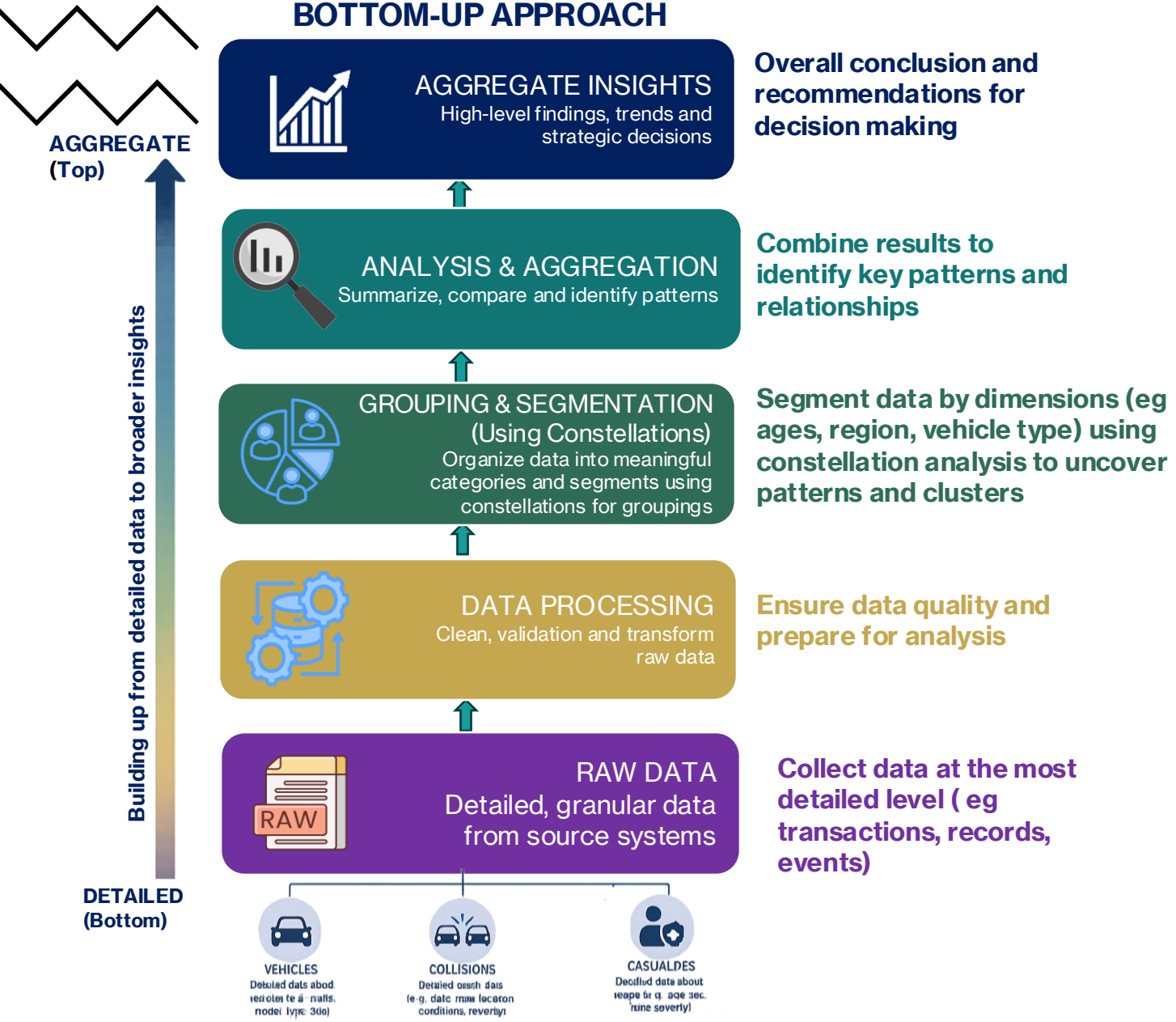
Data transformation was performed using R:

- Cleaning missing values
- Removing duplicates
- Formatting variables
- Creating new variables (work_and_personal_vehicles, engine_capacity_bands)
- Data Validity

Cleaned data was loaded into Hadoop for storage and analysis using HiveQL
Final insights were visualized using Tableau



Overview of the Data Warehouse Development



While the Top-Down Approach focuses on building a centralized, normalized enterprise data warehouse first, we have used the Kimball Method (Bottom-Up Approach).

In doing this, we have created several data marts(star schemas) consisting of multiple fact tables and connected through shared dimensions to form constellations in answering our business questions.

Advanced Data Management Applied



Why ETL?
- ETL is mostly considered for batch processing while ELT is mostly used for real time data
- GOV data contains possible inconsistencies, missing values, and formatting issues, ETL enables cleaning and standardisation in R before storage
- Improves storage efficiency and reduces query time



Why HDFS/Hadoop for storage?
- Handles large data effectively

Why Constellation Schema?
- Allows linking different but related business processes together (dim_time, collision_severities, casualties_severities)
- Using shared dimensions ensures consistent metrics across the different business questions
- Sharing dimension tables instead of recreating them for every fact table reduces the overall storage footprint of dimension data compared to maintaining multiple, independent star schemas.



Why Data Aggregation/joining done in Hive?
- Hive supports HiveQL; supports beginner friendly quering in aggregating and joining datasets
- Also optimized for large-scale batch processing



Why Data Cleaning and Validation in R?
- Contains different packages e.g tidyverse that runs large-scale batch processing in seconds
- Connects directly through the R function to pull the data from the gov.uk website without having it saved on the local system

Why Visualation in Tableau?
- Easily connects to multiple data sources (in this case optimised with Hive)
- Can create interactive visualisations of large datasets
- Beginner friendly with drag to screen tools

Screenshots of the Development Process

```
# Import Datasets
```{r}
dt_co <- fread("https://data.dft.gov.uk/road-accidents-safety-data/dft-road-casualty-statistics-collision-last-5-years.csv")

dt_ve <- fread("https://data.dft.gov.uk/road-accidents-safety-data/dft-road-casualty-statistics-vehicle-last-5-years.csv")

dt_ca <- fread("https://data.dft.gov.uk/road-accidents-safety-data/dft-road-casualty-statistics-casualty-last-5-years.csv")

```
```

The dataset was imported directly from the gov.uk website into R

```
#filter the data to only include final desired variables
```{r}
dt_co <- dt_co %>%
  select(
    collision_index,
    collision_severity,
    collision_date,
    light_conditions,
    urban_or_rural_area
    )

dt_ve <- dt_ve %>%
  select(
    collision_index,
    vehicle_reference,
    vehicle_type,
    sex_of_driver,
    age_band_of_driver,
    engine_capacity_bands,
    age_of_vehicle,
    generic_make_model,
    work_or_personal
    )

dt_ca <- dt_ca %>%
  select(
    collision_index,
    vehicle_reference,
    casualty_reference,
    casualty_severity
    )
```
```

The data was cleaned, validated and exported as a parquet file (Please see appendix for the complete R code.

Parquet is designed for "write-once, read-many (WORM)" scenarios and is much faster for queries requiring only a subset of columns.

```
#save final cleaned data to parquet format
```{r}
write_parquet(dt_co, 'collisions.parquet')
write_parquet(dt_ve, 'vehicles.parquet')
write_parquet(dt_ca, 'casualties.parquet')
```
```

Screenshots of the Development Process

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Student> scp -P 22 "C:\Users\Student\Desktop\rta_olap\Collisions.parquet" hadoop@localhost:/home/hadoop/rta_olap/input/
hadoop@localhost's password:
Collisions.parquet                                                                    100% 5097KB  16.7MB/s   00:00
PS C:\Users\Student>
PS C:\Users\Student> scp -P 22 "C:\Users\Student\Desktop\rta_olap\Vehicles.parquet" hadoop@localhost:/home/hadoop/rta_olap/input/
hadoop@localhost's password:
Permission denied, please try again.
hadoop@localhost's password:
Vehicles.parquet                                                                      100% 8349KB  22.1MB/s   00:00
PS C:\Users\Student>
PS C:\Users\Student> scp -P 22 "C:\Users\Student\Desktop\rta_olap\Casualties.parquet" hadoop@localhost:/home/hadoop/rta_olap/input/
hadoop@localhost's password:
Casualties.parquet                                                                    100% 6108KB  16.9MB/s   00:00
PS C:\Users\Student>
```

```
| vehicles.collision_index | vehicles.vehicle_reference | vehicles.vehicle_type | vehicles.sex_of_driver | vehicles.age_band_of_driver | vehicles.engine_capacity_bands | vehicles.age_of_vehicle | vehicles.generic_make_model | vehicles.work_or_personal |
+--------------------------+----------------------------+-----------------------+------------------------+-----------------------------+--------------------------------+-------------------------+-----------------------------+---------------------------+
| 2020210979534            | 2                          | 5                     | 1                      | 9                           | 26                             | 4                       | BMW F 800                   | 0                         |
| 2020830959800            | 1                          | 5                     | 1                      | 9                           | 26                             | 3                       | BMW F 800                   | 0                         |
| 2020360978858            | 2                          | 5                     | 1                      | 9                           | 26                             | 11                      | BMW F 800                   | 0                         |
| 2020430340711            | 1                          | 5                     | 1                      | 8                           | 26                             | 8                       | BMW F 800                   | 0                         |
| 2020810260297            | 1                          | 5                     | 1                      | 7                           | 26                             | 9                       | BMW F 800                   | 0                         |
| 2020810285121            | 2                          | 4                     | 1                      | 7                           | 26                             | 7                       | NULL                        | 0                         |
| 2020500930130            | 3                          | 5                     | 1                      | 4                           | 26                             | 14                      | BMW F 800                   | 0                         |
| 2020630835320            | 1                          | 5                     | 1                      | 10                          | 26                             | 4                       | BMW F 800                   | 0                         |
| 2020500978118            | 2                          | 5                     | 1                      | 8                           | 26                             | 7                       | BMW F 800                   | 0                         |
| 2020530982483            | 1                          | 5                     | 1                      | 10                          | 26                             | 2                       | BMW F 800                   | 0                         |
```

```
| casualties.collision_index | casualties.vehicle_reference | casualties.casualty_reference | casualties.casualty_severity |
+----------------------------+------------------------------+-------------------------------+------------------------------+
| 2020010280094              | 1                            | 1                             | 3                            |
| 202031D109620              | 1                            | 1                             | 2                            |
| 2020401083715              | 1                            | 1                             | 3                            |
| 2021201086330              | 1                            | 1                             | 3                            |
| 2021371050963              | 1                            | 1                             | 3                            |
| 2022161255871              | 1                            | 1                             | 3                            |
| 2023201345434              | 1                            | 1                             | 3                            |
| 2020401434010              | 1                            | 1                             | 2                            |
| 2020140992628              | 2                            | 1                             | 3                            |
| 202031D139820              | 2                            | 1                             | 3                            |
```

The exported parquet file was copied into Ubuntu Linux and then loaded into the Hadoop warehouse.
Transformation of data and joining of the Tables were also done here. (Please see appendix)

```
hadoop@hallam:~$ hdfs dfs -ls /rta_olap/input/collisions
hdfs dfs -ls /rta_olap/input/vehicles
hdfs dfs -ls /rta_olap/input/casualties
Found 1 items
-rw-r--r--   1 hadoop supergroup    5219720 2026-05-02 16:06 /rta_olap/input/collisions/Collisions.parquet
Found 1 items
-rw-r--r--   1 hadoop supergroup    8549661 2026-05-02 16:06 /rta_olap/input/vehicles/Vehicles.parquet
Found 1 items
-rw-r--r--   1 hadoop supergroup    6254309 2026-05-02 16:07 /rta_olap/input/casualties/Casualties.parquet
hadoop@hallam:~$
```

```
| collisions.collision_index | collisions.collision_severity | collisions.collision_date | collisions.light_conditions | collisions.urban_or_rural_area |
+----------------------------+-------------------------------+---------------------------+-----------------------------+--------------------------------+
| 2021170H10421              | 3                             | 2021-05-22                | 6                           | 2                              |
| 2021170H11231              | 3                             | 2021-10-20                | 1                           | 1                              |
| 2020170M11750              | 3                             | 2020-12-01                | 4                           | 1                              |
| 2021170M31761              | 3                             | 2021-12-09                | 4                           | 2                              |
| 2021170S10441              | 3                             | 2021-04-12                | 1                           | 2                              |
| 2022170S10952              | 3                             | 2022-07-05                | 1                           | 1                              |
| 2021111119419              | 2                             | 2021-12-09                | 1                           | 2                              |
| 2022111155778              | 3                             | 2022-03-18                | 1                           | 1                              |
| 2022111200178              | 2                             | 2022-06-22                | 1                           | 1                              |
| 2020070754597              | 3                             | 2020-12-12                | 1                           | 2                              |
```

Evidence of data quality checks

COLLISIONS

VARIABLE	RAW	DUPLICATE	MISSING	LOADED
collision_index	503,475	0	60	503,388
collision_severity	503,475	0	60	503,388
collision_date	503,475	0	60	503,388
light_conditions	503,475	0	79	503,388
urban_or_rural_area	503,475	0	68	503,388

The collisions dataset was scanned for duplicates to be removed (even though duplicates were non-existent). Total missing columns were scanned and 87 rows were removed(please see R code in appendices).

CASUALTIES

VARIABLE	RAW	DUPLICATE	MISSING	LOADED
collision_index	640,522	0	0	640,522
vehicle_reference	640,522	0	0	640,522
casualty_reference	640,522	0	0	640,522
casualty_severity	640,522	0	0	640,522

The casualties dataset was also checked for duplicates and missing values before loading.

VEHICLES

VARIABLE	RAW	DUPLICATE	SPECIFIC MISSING ROWS	LOADED
collision_index	920,692	0	141,382	779,310
vehicle_reference	920,692	0	141,382	779,310
vehicle_type	920,692	0	141,382	779,310
sex_of_driver	920,692	0	141,382	779,310
age_band_of_driver	920,692	0	141,382	779,310
engine_capacity_bands	920,692	0	141,382	779,310
age_of_vehicle	920,692	0	141,382	779,310
generic_make_model	920,692	0	141,382	779,310
work_or_personal	920,692	0	141,382	779,310

The vehicles dataset was also scanned for duplicates, in this case some of the observations were filled while others were removed.

Evidence of data quality checks

#Checking for duplicate rows

```
duplicate_row_num_and_perc <- function(dt) {
  total_rows <- nrow(dt)
  duplicate_rows <- sum(duplicated(dt))
  perc <- (duplicate_rows / total_rows) * 100
  cat("Number of duplicate rows for Casualties:", duplicate_rows, "\n")
  cat("Percentage of duplicate rows for Casualties:", perc, "%\n")
}

cat('\n', 'collisions dataset:', '\n')
```

```
##
##  collisions dataset:
```

```
duplicate_row_num_and_perc(dt_co)
```

```
## Number of duplicate rows for Casualties: 0
## Percentage of duplicate rows for Casualties: 0 %
```

#Remove Duplicates

```
dt_unique_co <- unique(dt_co)
dt_unique_ve <- unique(dt_ve)
dt_unique_ca <- unique(dt_ca)
```

#Check uniqueness

```
check_uniqueness_co <- nrow(dt_unique_co) == uniqueN(dt_co)
check_uniqueness_ve <- nrow(dt_unique_ve) == uniqueN(dt_ve)
check_uniqueness_ca <- nrow(dt_unique_ca) == uniqueN(dt_ca)

check_uniqueness_co
```

```
## [1] TRUE
```

```
check_uniqueness_ve
```

```
## [1] TRUE
```

```
check_uniqueness_ca
```

```
## [1] TRUE
```

```
lapply(dt_co, class)
```

```
## $collision_index
## [1] "character"
##
## $collision_year
## [1] "integer"
##
## $collision_ref_no
## [1] "character"
```

We checked for duplicate rows in the datasets, removed them, checked for uniqueness and also the class of each variables to ensure they are in their appropriate format.

#format fixing

```
co_time <- c("date")
dt_co[, (co_time) := lapply(co_time, function(x) as.Date(get(x), format = "%d/%m/%Y"))]
```

```
char_2_time_vars <- c("time")

append_seconds <- function(x) {
x <- as.character(x)
ifelse(is.na(x), x, paste0(x, ":00"))
}

hhmm_or_empty <- function(x) {
x[!is.na(x) & x == "NA"] <- NA
wrong_entry <- !is.na(x) & x != "" & !grepl("^\\d{2}:\\d{2}$", x)
x[wrong_entry] <- NA
x
}

dt_co[, (char_2_time_vars) := lapply(mget(char_2_time_vars), hhmm_or_empty)]

dt_co[, (char_2_time_vars) := lapply(mget(char_2_time_vars), append_seconds)]

dt_co[, (char_2_time_vars) := lapply(mget(char_2_time_vars), hms::as_hms)]
```

```
co_chars <- c("collision_index", "collision_ref_no")
dt_co[, (co_chars) := lapply(co_chars, function(x) as.character(get(x)))]
```

Evidence of data quality checks

#Check uniqueness

```{r}
check_uniqueness_co <- nrow(dt_unique_co) == uniqueN(dt_co)
check_uniqueness_ve <- nrow(dt_unique_ve) == uniqueN(dt_ve)
check_uniqueness_ca <- nrow(dt_unique_ca) == uniqueN(dt_ca)

check_uniqueness_co
check_uniqueness_ve
check_uniqueness_ca
```

A uniqueness check was performed to ensure there are **no duplicate records** in each dataset.
The number of rows was compared with the number of **unique records using uniqueN().**
That was applied to Collision dataset, vehicle dataset and casualty dataset.
The results returned **TRUE**, confirming that all records are unique.

#format fixing

```{r}
co_time <- c("date")
dt_co[, (co_time) := lapply(co_time, function(x) as.Date(get(x), format = "%d/%m/%Y"))]
```

The date field was originally stored as a **character (text) format,** so the date column was converted into a **proper Date format** to ensure consistency.

##Data Class Check

```{r}
lapply(dt_co, class)
```

```{r}
lapply(dt_ve, class)
```

```{r}
lapply(dt_ca, class)
```

A data class check was performed on all datasets such as Collision (dt_co), Vehicle (dt_ve), Casualty (dt_ca).
The lapply(..., class) function was used to identify the data type of each column.

Evidence of data quality checks

```r
#clean missing date values.
```{r}
dt_co <- dt_co[
  complete.cases(dt_co[ , c(
    'collision_index',
    'collision_severity',
    'date',
    'light_conditions',
    'urban_or_rural_area'
    )]), ]

summary(dt_co %>%
          select(
            collision_index,
            collision_severity,
            date,
            light_conditions,
            urban_or_rural_area))
```
```

```
 collision_index    collision_severity      date           light_conditions urban_or_rural_area
 Length:503388      Min.   :1.000      Min.   :2020-01-01   Min.   :1.000    Min.   :1.000
 Class :character   1st Qu.:3.000      1st Qu.:2021-05-30   1st Qu.:1.000    1st Qu.:1.000
 Mode  :character   Median :3.000      Median :2022-07-25   Median :1.000    Median :1.000
                    Mean   :2.752      Mean   :2022-07-23   Mean   :2.024    Mean   :1.325
                    3rd Qu.:3.000      3rd Qu.:2023-10-06   3rd Qu.:4.000    3rd Qu.:2.000
                    Max.   :3.000      Max.   :2024-12-31   Max.   :7.000    Max.   :2.000
```

Collision dataset missing values were identified in key variables; collision_index, collision_severity, date, light_conditions, urban_or_rural_area.
These fields were removed using "complete.cases()" function.

```r
#CASUALTIES DATASET CLEANING
```{r}
dt_ca <- dt_ca[
  complete.cases(dt_ca[ , c(
    'collision_index',
    'vehicle_reference',
    'casualty_reference',
    'casualty_severity'
    )]), ]

summary(dt_ca %>%
          select(
            collision_index,
            vehicle_reference,
            casualty_reference,
            casualty_severity
            ))
```
```

```
 collision_index    vehicle_reference casualty_reference casualty_severity
 Length:640522      Min.   :  1.000   Min.   :  1.000    Min.   :1.000
 Class :character   1st Qu.:  1.000   1st Qu.:  1.000    1st Qu.:3.000
 Mode  :character   Median :  1.000   Median :  1.000    Median :3.000
                    Mean   :  1.453   Mean   :  1.348    Mean   :2.786
                    3rd Qu.:  2.000   3rd Qu.:  1.000    3rd Qu.:3.000
                    Max.   :999.000   Max.   :999.000    Max.   :3.000
```

Casualities dataset missing values were identified in key variables collision_index, vehicle_reference, casualty_reference, casualty_severity. These fields were removed using "complete.cases()" function.

```r
#VEHICLES DATASET CLEANING
```{r}
ebike_type_change <- function(engine_capacity_cc) {
  ifelse(engine_capacity_cc <= 50, 2,
         ifelse(engine_capacity_cc <= 125, 3,
                ifelse(engine_capacity_cc <= 500, 4, 5)
      ))
}

dt_ve[vehicle_type == 97 & !is.na(engine_capacity_cc),
vehicle_type := ebike_type_change(engine_capacity_cc)]

cat('\n', 'Dimentions after fixing rows where vehicle type is a bike with unknown cc, but have an engine cc value:', '\n')
dim(dt_ve)
```
```

```
 Dimentions after fixing rows where vehicle type is a bike with unknown cc, but have an engine cc value:
[1] 920692     32
```

For Vehicle dataset, a lot of transformations were done. Some include categorizing and mapping engine size to ebike (If a bike has an unknown type but we know its engine size, classify it properly using engine capacity).
We also removed rows where vehicle type is unknown and also doesnt provide an engine capacity, weight/size and engine capacity (Please see R code appendix).
Some other rows were also removed (please see appendix)

Evidence of data quality checks

```
# NA counts per column
```{r}
colSums(is.na(dt_ve))
```
```{r}
colSums(is.na(dt_co))
```
```{r}
colSums(is.na(dt_ca))
```
```

Uses colSums(is.na()) on each dataset to count missing values per column, providing a quick completeness check after the placeholder recoding.

```
# Replace placeholder/missing codes with NA
```{r}
dt_co[dt_co == -1] <- NA
dt_co[dt_co == 99] <- NA
dt_co[dt_co$urban_or_rural_area == 3] <- NA
dt_co[dt_co == ""] <- NA
```

```{r}
dt_ve[dt_ve == -1] <- NA
dt_ve[dt_ve == 99] <- NA
dt_ve[dt_ve == ""] <- NA
dt_ve[generic_make_model == "<NA>", generic_make_model := NA]
dt_ve[generic_make_model == "MAKE AND MODEL REDACTED", generic_make_model := NA]
```
```

Converts values -1, 99, 3, , to NA, and MAKE AND MODEL REDACTED into proper NA values across the collisions and vehicles datasets, ensuring missing data is represented consistently for analysis.

```
#create variable for work/personal vehicle

dt_ve <- dt_ve %>%
  mutate(work_or_personal = case_when(
    vehicle_type %in% c(8, 11, 17, 18, 20, 21) |
      journey_purpose_of_driver %in% c(1, 8, 7) ~ 1,
    TRUE ~ 0
    ))
```

A variable to differentiate work and personal vehicles was also created from the journey purpose of the driver

We also banded (ranged) the engine capacity to make it easier to analyze

```
#create banded engine capacity variable.

eng_banding <- function(engine_capacity_cc){
  band_limits <- c(0, 25, 50, 75, 100, 125, 150, 175, 200, 225,
                   250, 275, 300, 325, 350, 375, 400, 425, 450, 475,
                   500, 550, 600, 650, 700, 750, 800, 850, 900, 950,
                   1100, 1200, 1300, 1400, 1500, 1600, 1700, 1800, 1900, 2000,
                   2100, 2200, 2300, 2400, 2500, 2600, 2700, 2800, 2900, 3000,
                   3100, 3200, 3300, 3400, 3500, 3600, 3700, 3800, 3900, 4000,
                   4100, 4200, 4300, 4400, 4500, 4600, 4700, 4800, 4900, 5000,
                   5250, 5500, 5750, 6000, 6250, 6500, 6750, 7000, 7250, 8250,
                   8500, 8750, 9000, 9250, 9500, 9750, 10000, 10500, 11000, 11500,
                   12000, 12500, 13000, 13500, 14000, 14500, 15000, 15600, 16000, 16500,
                   17000, 17500, 18000, 18500, 19000, 19500, 20000, 21000, 22000, 23000,
                   24000, 25000, 26000, 27000, 28000, 28500, 29000, 30000, 31000, 32000,
                   33000, 34000, 35000, 40000, 45000, 50000, Inf)

  band <- cut(engine_capacity_cc,
              breaks = band_limits,
              labels = FALSE,
              right = FALSE
              )
}

dt_ve <- dt_ve %>%
  mutate(engine_capacity_bands = eng_banding(engine_capacity_cc))
```

Evidence of data quality checks

#check timliness

```
check_timeliness <- function(DT) {
stopifnot(is.data.table(DT))
yr <- as.integer(format(DT$date, "%Y"))
mo <- as.integer(format(DT$date, "%m"))
ok <- all(!is.na(yr) & yr %in% c(2020:2024)) &&
  all(!is.na(mo) & mo %in% c(1L, 2L, 3L, 4L, 5L, 6L, 7L, 8L, 9L, 10L, 11L, 12L))
cat(if (ok) "TRUE\n" else "FALSE\n")
}
check_timeliness(dt_co)
```

This function validates that all records contain valid dates within the specified time range (2020–2024) and ensures that month values are correctly formatted

```
## TRUE
```

#check again for duplicates using natural keys

```
key_cols <- c("collision_index", "vehicle_reference", "casualty_reference")
# Duplicate rate
n_total <- nrow(dt_ca)
n_unique <- uniqueN(dt_ca, by = key_cols)
dup_rows <- n_total - n_unique
dup_rate <- 100 * dup_rows / n_total
cat(sprintf("Duplicates: %d (%.2f%%) based on key: %s\n",
dup_rows, dup_rate, paste(key_cols, collapse = ", ")))
```

This evaluates the data quality by identifying duplicate records, calculating both the number and percentage of duplicates to ensure dataset integrity before analysis.

#rename date variable to avoid confusion in HiveQL

```
dt_co <- dt_co %>%
  rename(collision_date = date)
```

We had to rename the variable date to collision_date because "date" is a reserved function in HiveQL.

#referential integrity uniqueness check

```
# NA checks
sapply(dt_co, function(x) sum(is.na(x)))
```

The function above simply answers the question "How many missing values does each column have?" Same function was applied to the vehicle, and casualties dataset (please refer to the R code in appendix)

```
# type of data checks
sapply(dt_co, typeof)
```

The function above prints out the data type of each variables in the dataset. Same function was applied to the vehicle, and casualties dataset (please refer to the R code in appendix)

Evidence of data quality checks

```
# Uniqueness checks (primary key candidates)
collisions_unique_ok <- length(unique(dt_co$collision_index)) == nrow(dt_co)
vehicles_unique_ok <- length(unique(paste(dt_ve$collision_index, dt_ve$vehicle_reference, sep=""))) == nrow(dt_ve)
casualties_unique_ok <- length(unique(paste(dt_ca$collision_index, dt_ca$casualty_reference, sep=""))) == nrow(dt_ca)

cat("Are all primary keys codes unique? ", collisions_unique_ok, "\n")
cat("Are all vehicle composite keys codes unique? ", vehicles_unique_ok, "\n")
cat("Are all casualty composite keys codes unique? ", casualties_unique_ok, "\n")

# How many duplicates + which codes (if any)
dup_co_n <- sum(duplicated(dt_co$collision_index))
dup_ve_n <- sum(duplicated(paste(dt_ve$collision_index, dt_ve$vehicle_reference, sep="")))
dup_ca_n <- sum(duplicated(paste(dt_ca$collision_index, dt_ca$casualty_reference, sep="")))

cat("Duplicate collision codes:", dup_co_n, "\n")
cat("Duplicate vehicle codes: ", dup_ve_n, "\n")
cat("Duplicate casualties codes: ", dup_ca_n, "\n")

dup_co_codes <- sort(unique(dt_co$collision_index[duplicated(dt_co$collision_index)]))
dup_ve_codes <- sort(unique(paste(dt_ve$collision_index, dt_ve$vehicle_reference, sep="")[duplicated(paste(dt_ve$collision_index,
dt_ve$vehicle_reference, sep=""))]))
dup_ca_codes <- sort(unique(paste(dt_ca$collision_index, dt_ca$casualty_reference, sep="")[duplicated(paste(dt_ca$collision_index,
dt_ca$casualty_reference, sep=""))]))

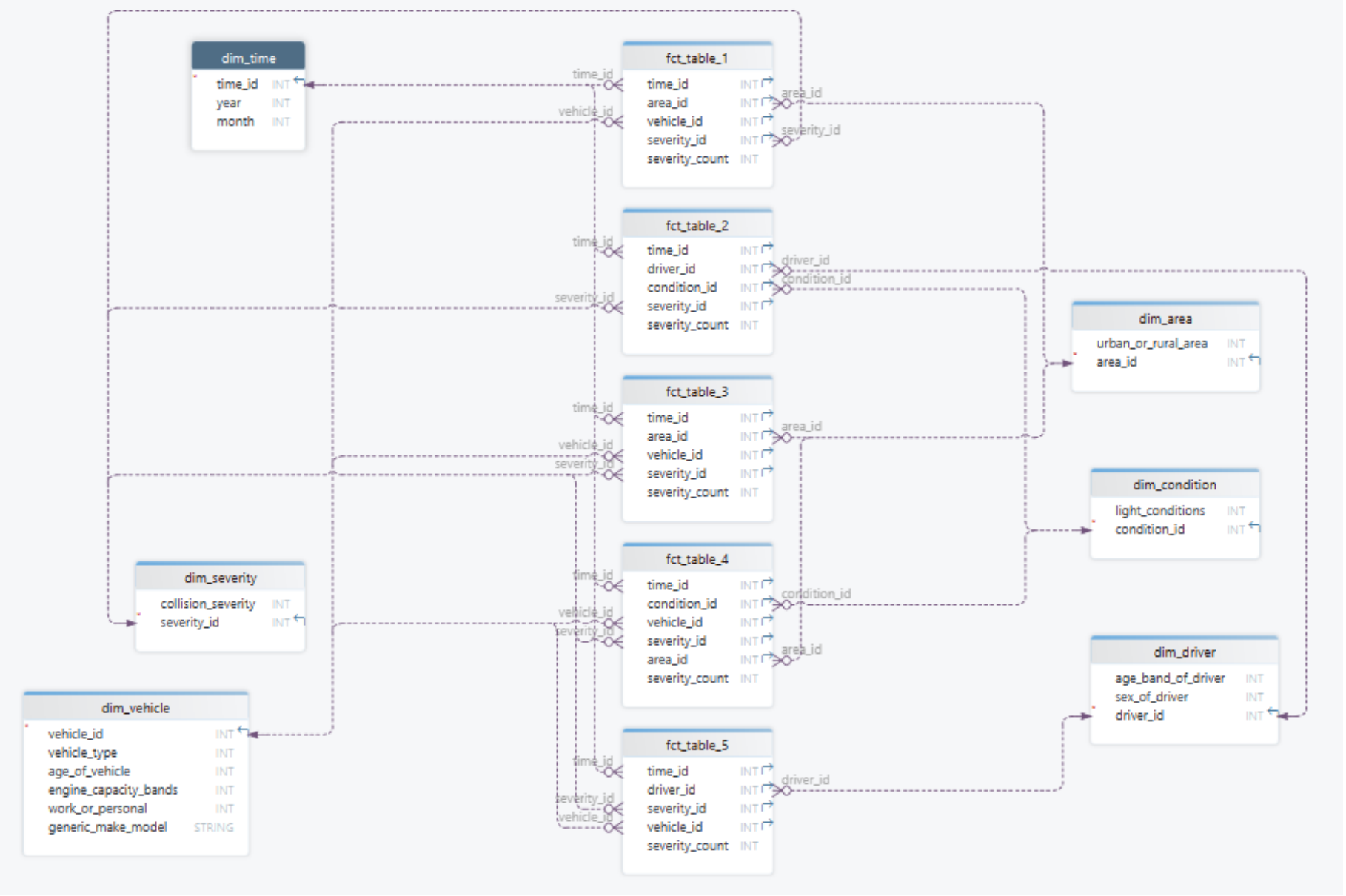
print(dup_co_codes)
print(dup_ve_codes)
print(dup_ca_codes)
```

We checked the uniqueness of the primary and composite keys.

We also checked for duplicates for the collision, vehicle and casualties codes and found that none exist.



Evidence of the developed data mart



This figure shows the relational diagram of the constellation drawn in DbSchema with all five fact tables and the respective dimensions.

The foreign keys inside the fact tables will collect data by its relationship with the primary keys in the dimension tables where the data is stored.

The fact tables each contain one metric variable "severity count" and will calculate the sum of collisions for each severity classification inside each fact table.

This is a one to many relationship because a single dimension can relate to multiple fact tables.

Evidence of data validity testing conducted

#FORMAT VALIDITY CHECK #collisions

```
col_class <- sapply(dt_co, function(x) paste(class(x), collapse = ","))
col_type <- sapply(dt_co, typeof)


groups <- list(
  integer = c("collision_year", "vehicle_reference", "vehicle_type",
              "towing_and_articulation", "vehicle_manoeuvre_historic",
              "vehicle_manoeuvre", "vehicle_direction_from",
              "vehicle_direction_to", "vehicle_location_restricted_lane_historic",
              "vehicle_location_restricted_lane", "junction_location",
              "skidding_and_overturning", "hit_object_in_carriageway",
              "vehicle_leaving_carriageway", "hit_object_off_carriageway",
              "first_point_of_impact", "vehicle_left_hand_drive",
              "journey_purpose_of_driver_historic", "journey_purpose_of_driver",
              "sex_of_driver", "age_of_driver", "age_band_of_driver",
              "engine_capacity_cc", "propulsion_code", "age_of_vehicle",
              "driver_imd_decile","escooter_flag","driver_distance_banding"),
  Date = "collision_date",
  hms = c("time"),
  numeric =
    c("collision_adusted_severity_serious","collision_adusted_severity_slight",
      "longitude", "latitude"),
  character = c("collision_index","collision_ref_no",
                "local_authority_ons_district","local_authority_highway",
                "local_authority_highway_current", "lsoa_of_accident_location")
)

preds <- list(
  integer = is.integer,
  Date = \(x) "Date" %in% class(x),
  hms = hms::is_hms,
  numeric = is.numeric,
  character = is.character
```

#casualties

```{r}
col_class <- sapply(dt_ca, function(x) paste(class(x), collapse = ","))
col_type <- sapply(dt_ca, typeof)


groups <- list(
  integer = c("collision_year", "vehicle_reference", "casualty_reference",
              "casualty_class",  "sex_of_casualty", "age_of_casualty",
              "age_band_of_casualty", "casualty_severity", "pedestrian_location",
              "pedestrian_movement",  "car_passenger", "bus_or_coach_passenger",
              "pedestrian_road_maintenance_worker", "casualty_type",
              "casualty_imd_decile", "enhanced_casualty_severity",
              "casualty_injury_based",  "casualty_distance_banding"),

  numeric =
    c("casualty_adusted_severity_serious","casualty_adusted_severity_slight"),
  character = "lsoa_of_casualty"
)

preds <- list(
  integer = is.integer,
  numeric = is.numeric,
  character = is.character
)
```

#vehicles

```{r}
col_class <- sapply(dt_ve, function(x) paste(class(x), collapse = ","))
col_type <- sapply(dt_ve, typeof)


groups <- list(
  integer = c("collision_year",
              "vehicle_reference", "vehicle_type", "towing_and_articulation",
              "vehicle_manoeuvre_historic", "vehicle_manoeuvre",
              "vehicle_direction_from", "vehicle_direction_to",
              "vehicle_location_restricted_lane_historic",
              "vehicle_location_restricted_lane", "junction_location",
              "skidding_and_overturning", "hit_object_in_carriageway",
              "vehicle_leaving_carriageway", "hit_object_off_carriageway",
              "first_point_of_impact", "vehicle_left_hand_drive",
              "journey_purpose_of_driver_historic", "journey_purpose_of_driver",
              "sex_of_driver", "age_of_driver", "age_band_of_driver",
              "engine_capacity_cc", "propulsion_code", 'age_of_vehicle',
              "driver_imd_decile", "escooter_flag", "driver_distance_banding"),

  character = c("generic_male_model", "lsoa_of_driver", 'collision_index',
                "collision_ref_no")
)

preds <- list(
```

Format validity checks were performed to ensure each variable matches the expected data type for collisions, causalities and vehicles.
Validation was applied across the collision dataset using with Integer, Date, Time (hms), Numeric, Character formats

Evidence of data validity testing conducted

#RANGE RULES

```
ve_nonull <- dt_ve[complete.cases(dt_ve), ]

v_collision_severity <- sum(!(dt_co$collision_severity %in% c(1L:3L)), na.rm = TRUE)
v_light_conditions <- sum(!(dt_co$light_conditions %in% c(1L:7L)), na.rm = TRUE)
v_urban_rural <- sum(!(dt_co$urban_or_rural_area %in% c(1L:3L)), na.rm = TRUE)

v_vehicle_type <- sum(!(ve_nonull$vehicle_type %in% c(1L:11L, 16L:23L, 90L, 97L:99L)), na.rm = TRUE)
v_vehicle_age <- sum(!(ve_nonull$age_of_vehicle >= 0))
v_engine_capacity_band <- sum(!(ve_nonull$engine_capacity_bands %in% c(1L:127L)), na.rm = TRUE)
v_work_or_personal <- sum(!(ve_nonull$work_or_personal %in% c(0L, 1L)), na.rm = TRUE)
v_driver_sex <- sum(!(ve_nonull$sex_of_driver %in% c(1L:3L)), na.rm = TRUE)
v_driver_age_band <- sum(!(ve_nonull$age_band_of_driver %in% c(1L:11L)), na.rm = TRUE)

v_casualty_severity <- sum(!(dt_ca$casualty_severity %in% c(1L:3L)), na.rm = TRUE)

v_any_co <- sum(
  !(
    dt_co$collision_severity %in% c(1L:3L) &
      dt_co$light_conditions %in% c(1L:7L) &
      dt_co$urban_or_rural_area %in% c(1L:3L)
  ), na.rm = TRUE )

v_any_ve <- sum(
  !(
    !is.na(ve_nonull$vehicle_type) %in% c(1L:11L, 16L:23L, 90L, 97L:99L) &
      ve_nonull$age_of_vehicle >= 0 &
      ve_nonull$engine_capacity_bands %in% c(1L:127L) &
      ve_nonull$work_or_personal %in% c(0L, 1L) &
      ve_nonull$sex_of_driver %in% c(1L:3L) &
      ve_nonull$age_band_of_driver %in% c(1L:11L)
  ), na.rm = TRUE )

v_any_ca <- sum(
  !(dt_ca$casualty_severity %in% c(1L:3L)), na.rm = TRUE
)

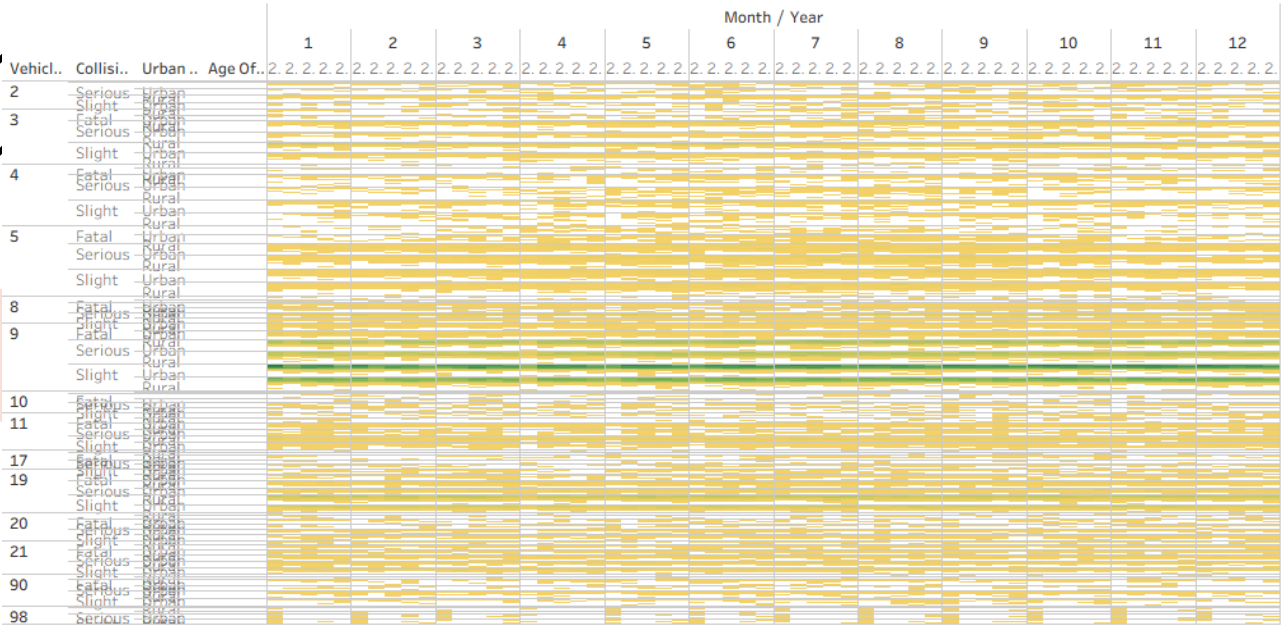
cat("invalid values for collision_severity: ", v_collision_severity, "\n")
```

Range rule validation was conducted to ensure that dataset values fall within the expected and valid ranges. The validation checks were applied across the three datase; Collision dataset, Vehicle dataset and Casualty dataset.
Key variables validated includes Collision severity, Light conditions , Urban/rural area, Vehicle type, Vehicle age, Engine capacity bands, Driver, sex, age bands and Casualty severity

Logical range conditions were defined using R validation rules and checked using conditional statements.

The results confirmed that the cleaned datasets contained valid and consistent values suitable for analysis

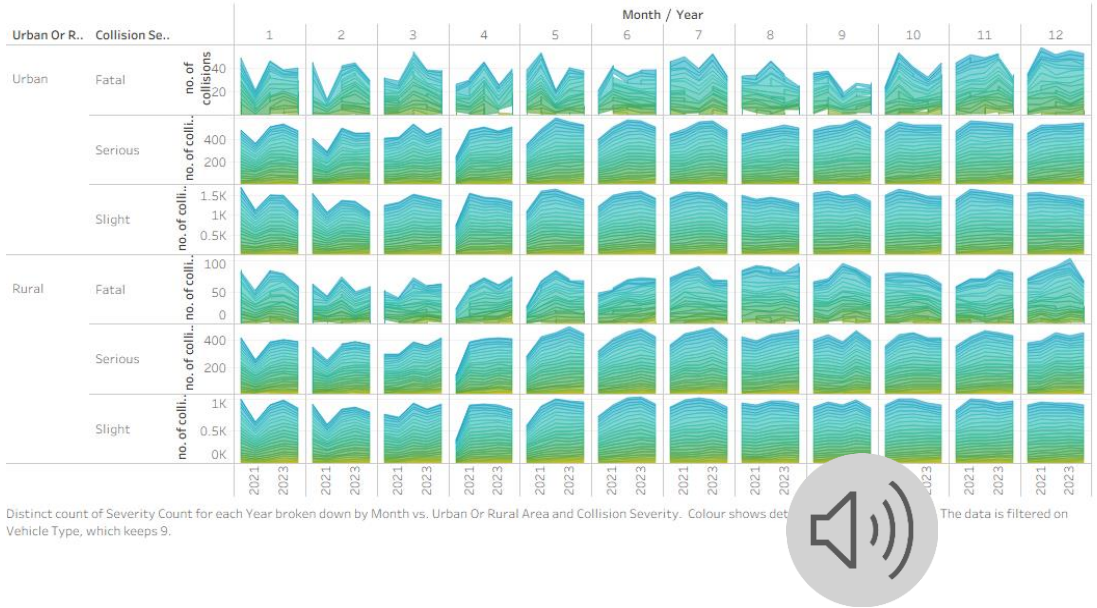
Sum of collision severities by vehicle type, vehicle age band, within an urban or rural area per month between 2020-2024



Distinct count of Severity Count (colour) broken down by Month and Year vs. Vehicle Type, Collision Severity, Urban Or Rural Area and Age Of Vehicle.

Distinct count of Severi..
1
129

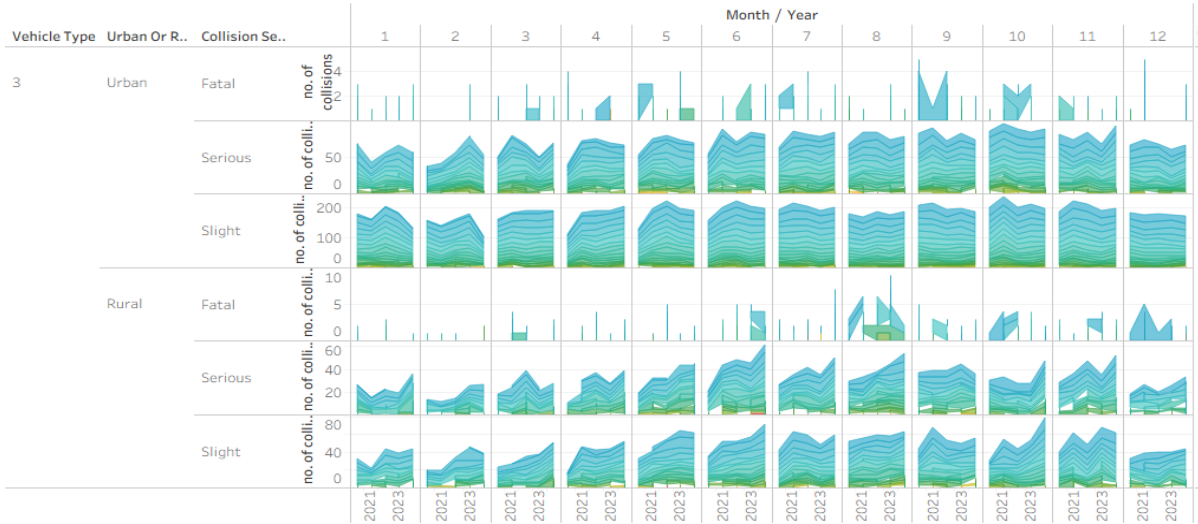
no. of car collisions by vehicle age, collision severity, urban or rural area and month between 2020-2024 in the UK



Distinct count of Severity Count for each Year broken down by Month vs. Urban Or Rural Area and Collision Severity. Colour shows det... The data is filtered on Vehicle Type, which keeps 9.

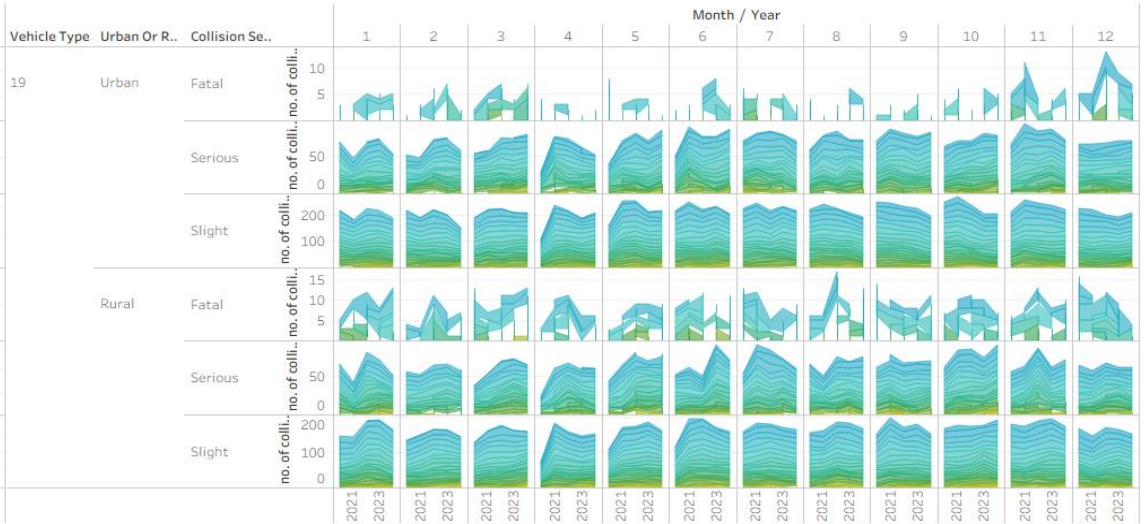
Age Of Vehicle
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 62 67 69 74 75 86 91

No. of collisions for motorcycles (125cc or under) by vehicle age, collision severity, urban or rural area and month between 2020-2024 in the UK



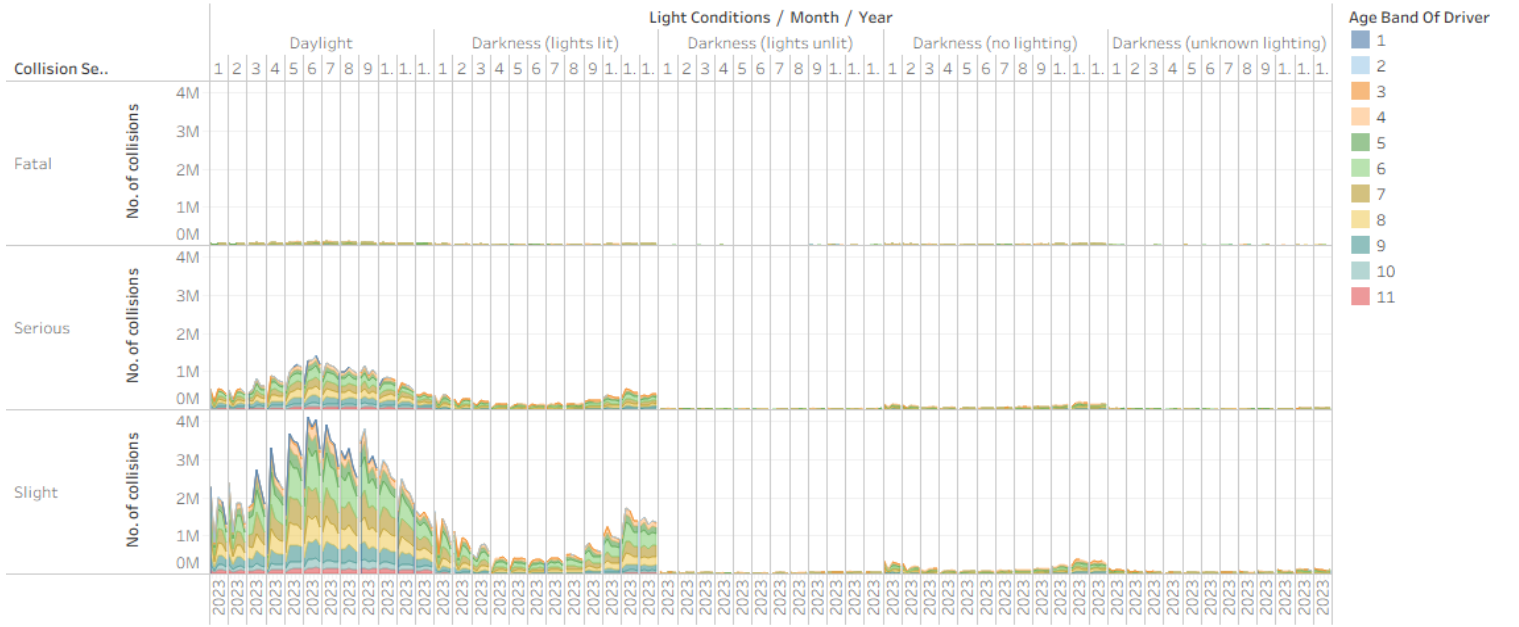
Distinct count of Severity Count for each Year broken down by Month vs. Vehicle Type, Urban Or Rural Area and Collision Severity. Colour shows details about Age Of Vehicle. The view is filtered on Vehicle Type, which keeps 3.

No. of collisions for vans/goods vehicles under 3.5 tonnes by vehicle age, collision severity, urban or rural area and month between 2020-2024 in the UK



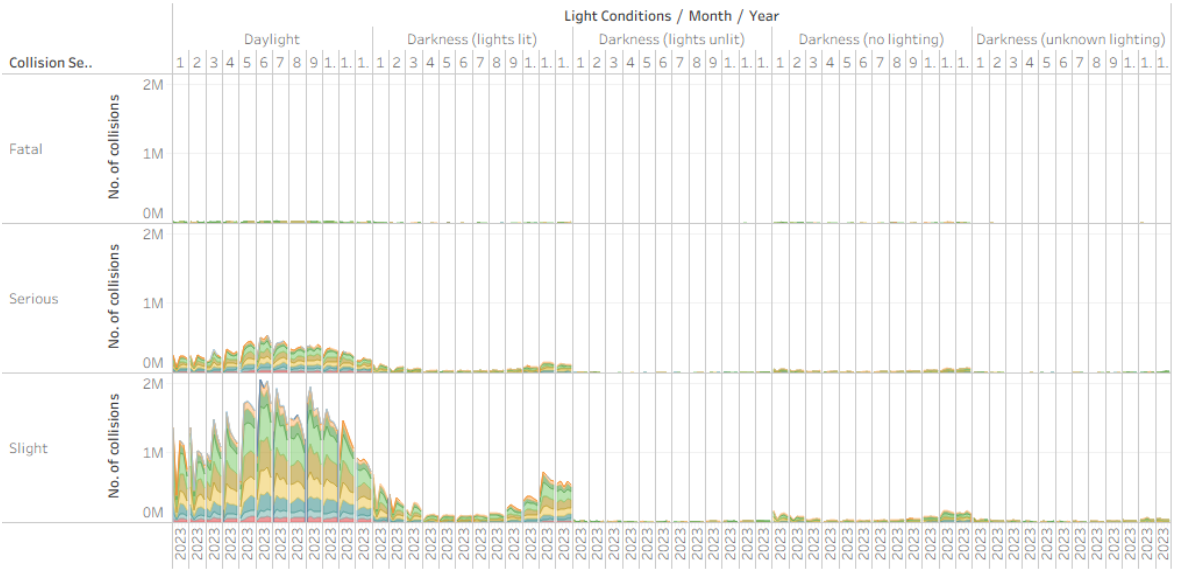
Distinct count of Severity Count for each Year broken down by Month vs. Vehicle Type, Urban Or Rural Area and Collision Severity. Colour shows details about Age Of Vehicle. The view is filtered on Vehicle Type, which keeps 19.

no. of collisions for males by collision severity, light conditions, driver age band and month between 2020-2024 in the UK



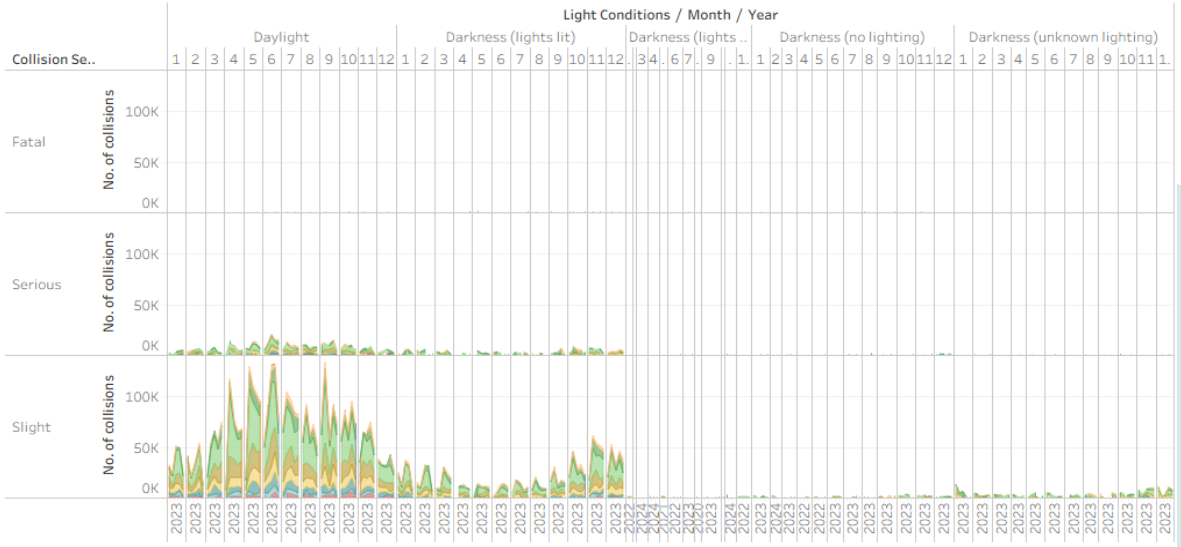
Sum of Severity Count (Fct Table 2) for each Year broken down by Light Conditions and Month vs. Collision Severity. Colour shows details about Age Band Of Driver. The data is filtered on Sex Of Driver, which keeps Male.

no. of collisions for females by collision severity, light conditions, driver age band and month between 2020-2024 in the UK



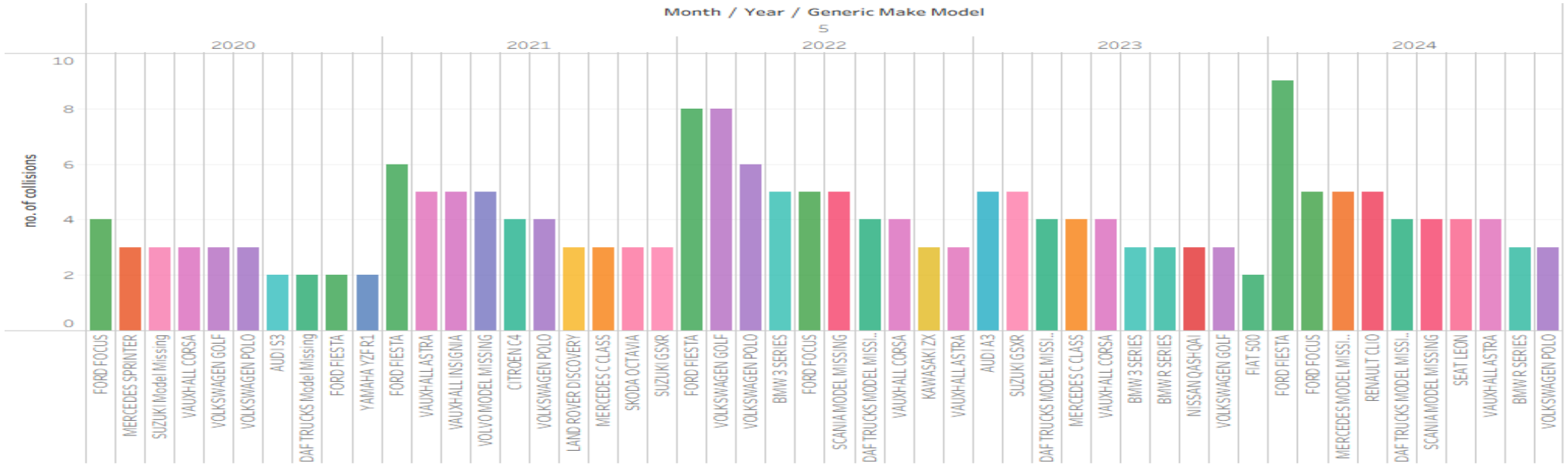
Sum of Severity Count (Fct Table 2) for each Year broken down by Light Conditions and Month vs. Collision Severity. Colour shows details about Age Band Of Driver. The data is filtered on Sex Of Driver, which keeps Female.

no. of collisions for drivers of unknown sex by collision severity, light conditions, driver age band and month between 2020-2024 in the UK



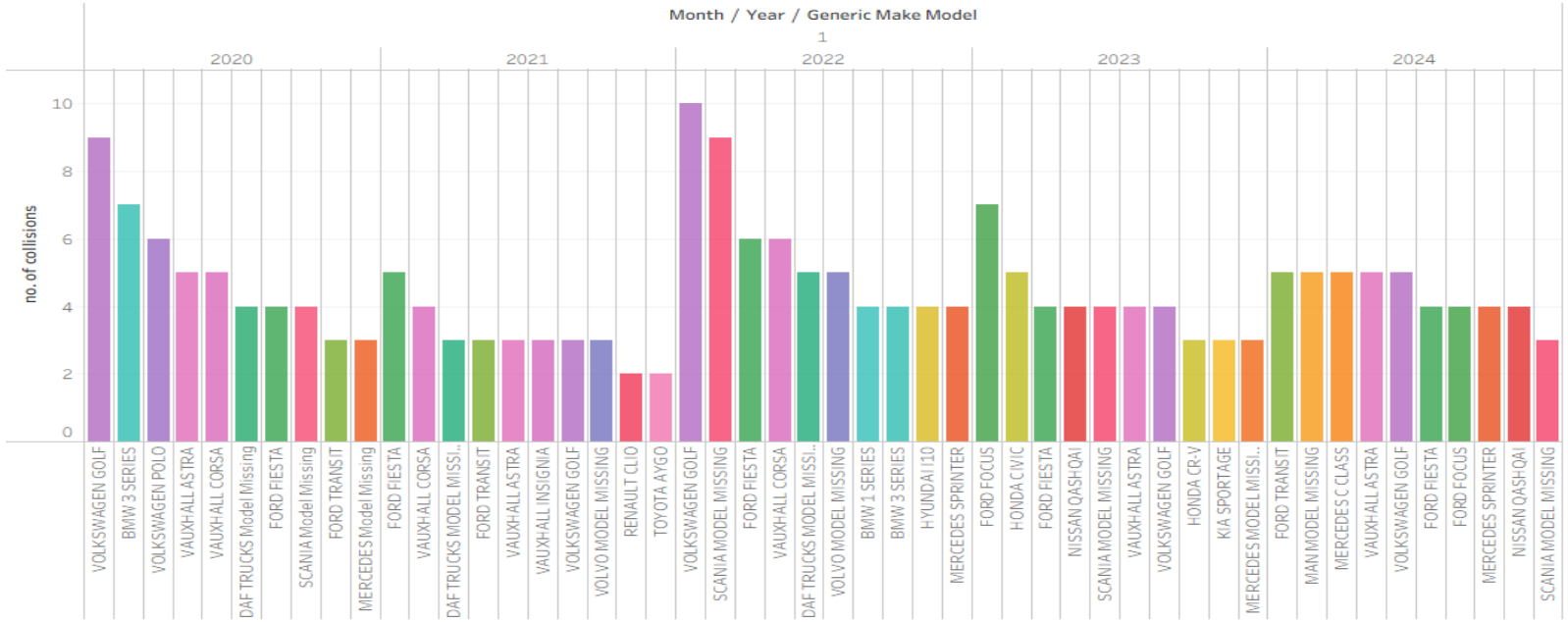
Sum of Severity Count (Fct Table 2) for each Year broken down by Light Conditions and Month vs. Collision Severity. Colour shows details about Age Band Of Driver. The data is filtered on Sex Of Driver, which keeps Unknown.

Most Dangerous vehicle makes and models by collision fatalities & month between 2020 and 2024 in the UK



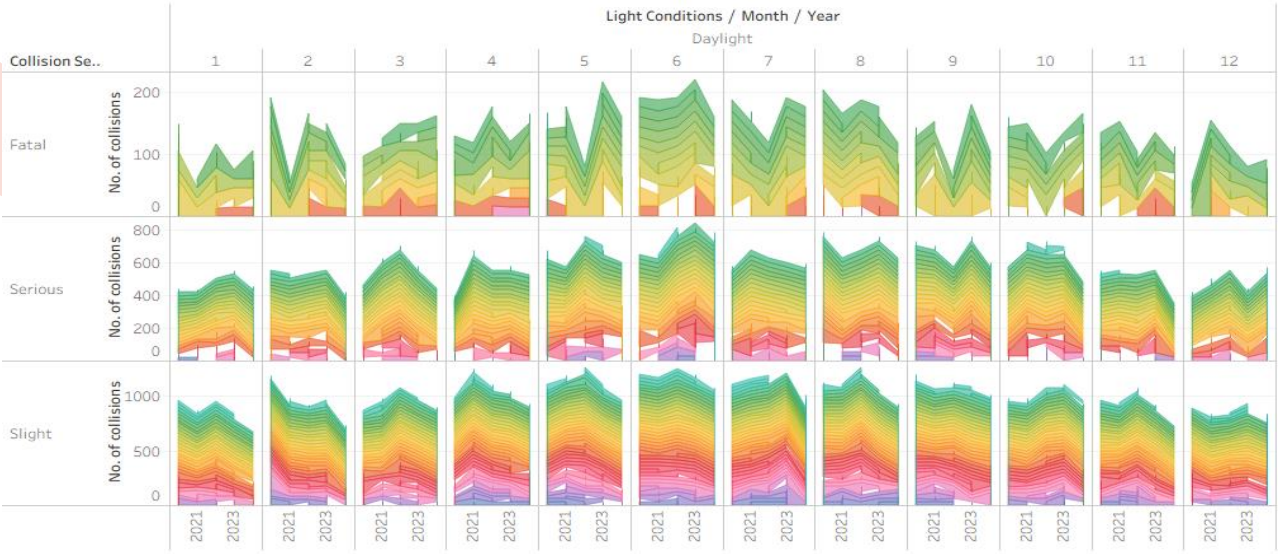
Count of Severity Count (Fct Table 3) for each indx broken down by Month, Year and Generic Make Model. Colour shows details about Generic Make Model. The data is filtered on Collision Severity and indx. The Collision Severity filter ranges from Fatal to Fatal. The indx filter keeps 10 members. The view is filtered on Month, which keeps 5.

Most Dangerous vehicle makes and models by collision fatalities & month between 2020 and 2024 in the UK



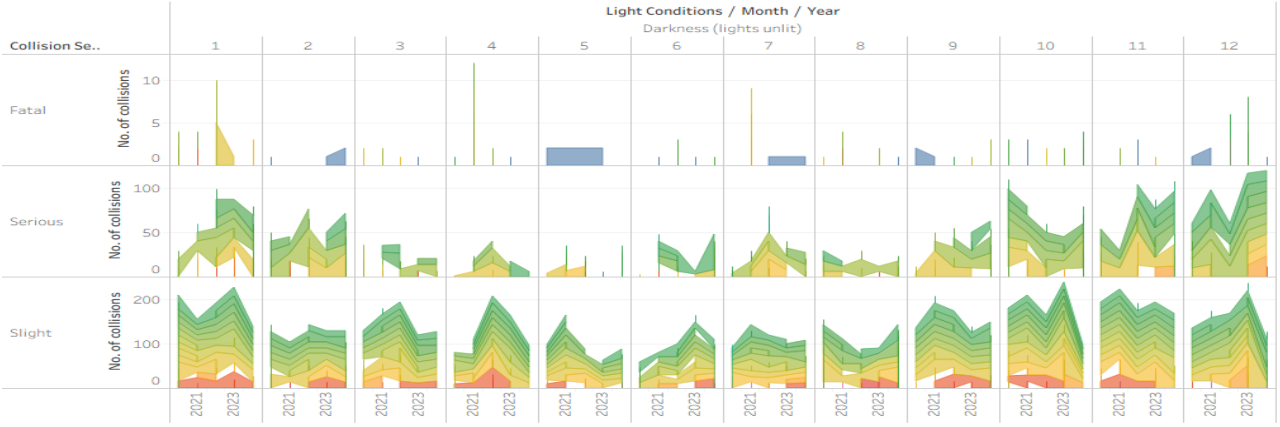
Count of Severity Count (Fct Table 3) for each indx broken down by Month, Year and Generic Make Model. Colour shows details about Generic Make Model. The data is filtered on Collision Severity and indx. The Collision Severity filter ranges from Fatal to Fatal. The indx filter keeps 10 members. The view is filtered on Month, which keeps 1.

no. of collisions for cars in daylight by collision severity, engine capacity and month between 2020-2024 in urban areas of the UK



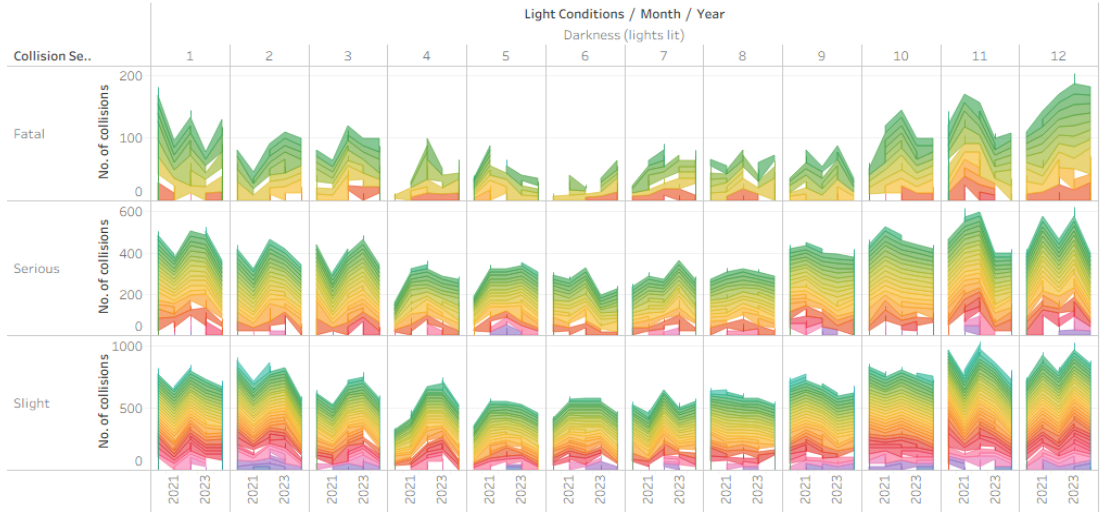
Distinct count of Severity Count (Fct Table 2) for each Year broken down by Light Conditions and Month vs. Collision Severity. Colour shows details about Engine Capacity Bands. The data is filtered on Vehicle Type and Urban Or Rural Area. The Vehicle Type filter keeps 9. The Urban Or Rural Area filter keeps Urban. The view is filtered on Light Conditions, which keeps Daylight.

no. of collisions for cars in darkness (lights unlit) by collision severity, engine capacity and month between 2020-2024 in urban areas of the UK



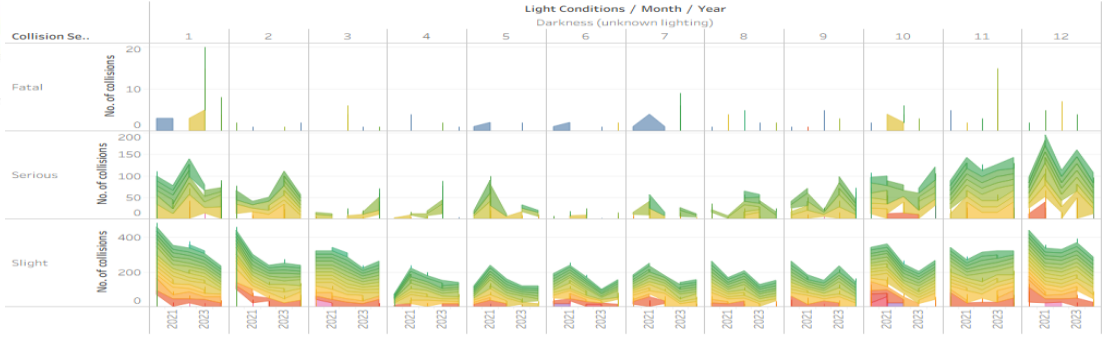
Distinct count of Severity Count (Fct Table 2) for each Year broken down by Light Conditions and Month vs. Collision Severity. Colour shows details about Engine Capacity Bands. The data is filtered on Vehicle Type and Urban Or Rural Area. The Vehicle Type filter keeps 9. The Urban Or Rural Area filter keeps Urban. The view is filtered on Light Conditions, which keeps Darkness (lights unlit).

no. of collisions for cars in darkness (lights lit) by collision severity, engine capacity and month between 2020-2024 in urban areas of the UK



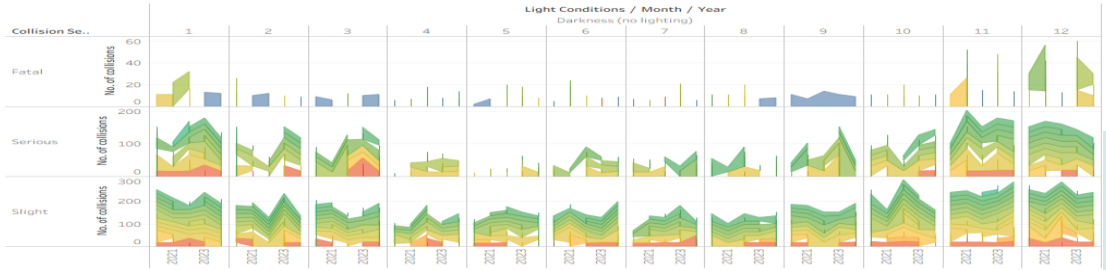
Distinct count of Severity Count (Fct Table 2) for each Year broken down by Light Conditions and Month vs. Collision Severity. Colour shows details about Engine Capacity Bands. The data is filtered on Vehicle Type and Urban Or Rural Area. The Vehicle Type filter keeps 9. The Urban Or Rural Area filter keeps Urban. The view is filtered on Light Conditions, which keeps Darkness (lights lit).

no. of collisions for cars in darkness (unknown lighting) by collision severity, engine capacity and month between 2020-2024 in urban areas of the UK

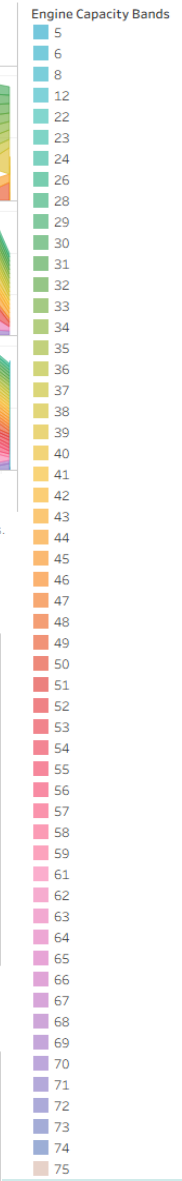


Distinct count of Severity Count (Fct Table 2) for each Year broken down by Light Conditions and Month vs. Collision Severity. Colour shows details about Engine Capacity Bands. The data is filtered on Vehicle Type and Urban Or Rural Area. The Vehicle Type filter keeps 9. The Urban Or Rural Area filter keeps Urban. The view is filtered on Light Conditions, which keeps Darkness (unknown lighting).

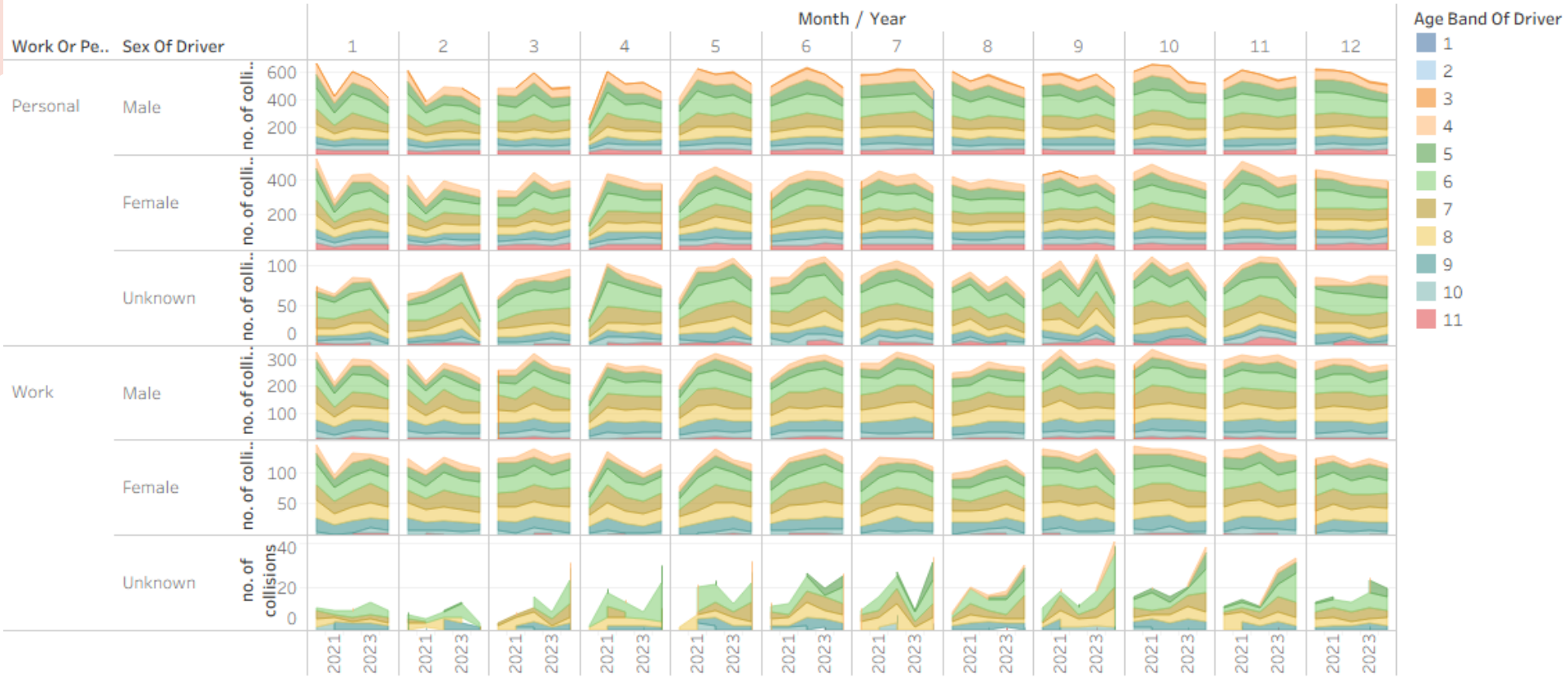
no. of collisions for cars in darkness (no lighting) by collision severity, engine capacity and month between 2020-2024 in urban areas of the UK



Distinct count of Severity Count (Fct Table 2) for each Year broken down by Light Conditions and Month vs. Collision Severity. Colour shows details about Engine Capacity Bands. The data is filtered on Vehicle Type and Urban Or Rural Area. The Vehicle Type filter keeps 9. The Urban Or Rural Area filter keeps Urban. The view is filtered on Light Conditions, which keeps Darkness (no lighting).



No. of collisions by severity, work or personal use vehicles, sex of driver, age group of driver and month from 2020 - 2024 in the UK



Distinct count of Severity Count (Fct Table 5) for each Year broken down by Month vs. Work Or Personal and Sex Of Driver. Colour shows details about Age Band Of Driver.

Final Recommendations

🚗 Q1 – Vehicle & Area Risk

- ❑ Higher premiums should be
  - ➢ Newer vehicles

  (Cars, Motorcycles <125cc, Vans <3.5 tonnes)
- ❑ Lower premiums should be
  - ➢ Vehicles older than 14 years
  - ➢ Motorcycles during winter based on RTA data.
- ❑Higher van risk in rural areas

👴 Q2 – Driver & Light Condition Risk

- ❑ Higher premiums should receive
  - ➢ Male drivers
  - ➢ Driver age groups 3–6
- ❑ Areas with limited lighting should receive increased premiums through winter.

🚚 Q3 – High-Risk Vehicle Models

- ❑ Common make/models should receive higher premium.
  - ➢ Volkswagen Golf
  - ➢ Ford Fiesta / Focus
  - ➢ DAF & Scania trucks
  - ➢ BMW Series 1–3
- ❑ Seasonal risks,
  - ➢ Trucks premiums raise in November
  - ➢ Motorcycles premiums raise in Spring/Summer.

Final Recommendations

⚙️ Q4 – Engine Capacity Risk

- ❑ Engine capacity bands 30–36 should receive increased premium recommendation due to higher collision frequency.
- ❑ Similar risk patterns observed from Q1 & Q2.

🚗 Q5 – Work Vehicle Risk

- ❑ Higher premiums should have
  - ➢ Male drivers
  - ➢ Driver age groups 3–6
- ❑ Lower risk happens
  - ➢ Work vehicles in age band 10 reduced insurance premium.

📌 Overall Recommendations

- ❑ Improve urban road safety monitoring
- ❑ Increase safety measures in low-light areas
- ❑ Support data-driven insurance and road safety decisions

AI Transparency, Appendix & References: Attached Separately



THANK YOU

Group 15